How-To Library

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Create and Use Standard Masking Rules in Test Data Management

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Abstract

In Test Data Management (TDM), you can create and assign standard data masking rules to mask sensitive data. This article describes how to create random and shuffle masking rules, assign the rules to target columns, and create and run a plan to mask the source data.

Supported Versions

- Test Data Management 9.5.1 to 10.1.0

Table of Contents

Overview. . . . 2
Scenario. . . . 2
Prerequisites. . . . 3
Step 1. Create a Numeric Random Masking Rule. . . . 3
Step 2. Create a Shuffle Masking Rule. . . . 4
Step 3. Add the Masking Rules to a Project. . . . 5
Step 4. Assign the Masking Rules to Target Columns. . . . 6
Step 5. Create a Data Masking Plan. . . . 7
Step 6. Generate and Run the Workflow. . . . 9
Sample Output. . . . 10

Overview

Create and use standard masking rules that apply built-in masking techniques to mask the source data.

You can select the available standard masking techniques based on the data type of the column.

You can use the following standard masking rules:

Random

Random masking produces random, non-repeatable results for the same source data. Random masking does not require a seed value. The results of random masking are non-deterministic. Use random masking to mask date, numeric, and string data types.

To mask numeric data types, you can configure blurring or provide a range of output values. When you define a range for numeric values, the PowerCenter Integration Service masks the source data with a value between the minimum and maximum values.

Shuffle

Shuffle masking masks the data in a column with data from the same column in another row of the table. Shuffle masking switches all the values for a column in a file or a database table.

You can configure shuffle masking to shuffle data randomly or you can configure shuffle masking to return repeatable results. Mask date, numeric, and string data types with shuffle masking.

Scenario

You work for an organization in the software testing department. You need to test a banking application that maintains customer details, such as account IDs, account types, minimum balance, and branch IDs. Before you test the

application, you need to mask the minimum balance and branch ID columns. Create and assign standard data masking rules to the source metadata, and create and run the plan to mask sensitive data.

To mask sensitive columns, perform the following tasks:

1. Create a numeric random masking rule.
2. Create a shuffle masking rule.
3. Add the masking rules to a project.
4. Assign the masking rules to target columns.
5. Create a data masking plan.
6. Generate and run the workflow.

The following image shows a sample source data:

ACCOUNT_ID	ACCOUNT_TYPE	MIN_BALANCE	BRANCH_ID
31926819	SAVINGS	16263	21211009
7099453	CURRENT	16782	19043
440116243	SALARY	15173	129007
12300695	SAVINGS	28813	539
1863000160	CURRENT	29514	BNBG9661
611111101800000000	SALARY	16603	762
12345678	SAVINGS	18477	200128
99999999	CURRENT	27190	800
260076545510730000	SALARY	29120	37040044
123456	SAVINGS	29561	40
0002324013AA	CURRENT	24576	22
11101001000	SALARY	21841	21000418
9400644750000	SAVINGS	21181	123456
435195001	CURRENT	26782	61
0011111000h79	SALARY	17944	2004101005
12345678951	SAVINGS	26482	NWBK601613
120000058984	CURRENT	23174	NWBK
0012345MTLCAST001S	SALARY	19378	110125

Prerequisites

Perform the following prerequisites tasks:

1. Create an Oracle database connection in Test Data Manager.
2. Create a project and import metadata from the Oracle database.

Step 1. Create a Numeric Random Masking Rule

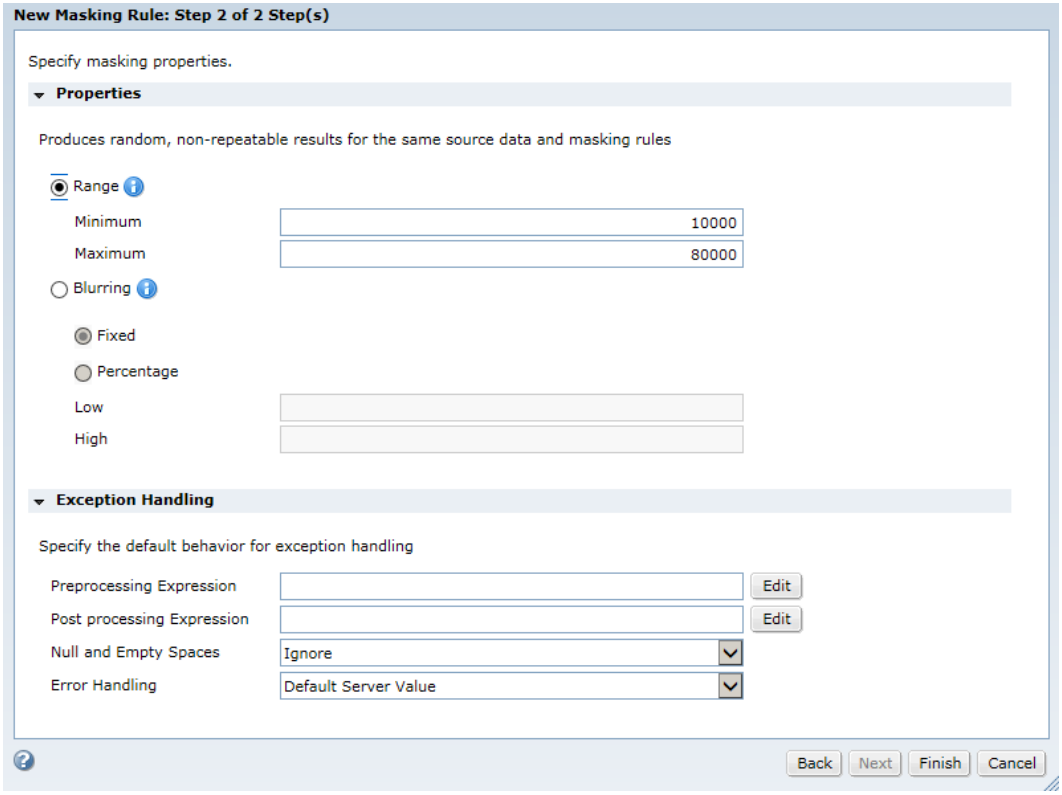
Create a numeric random masking rule and use the rule to mask the minimum account balance details of the customers. You can provide a range of values with which you want to mask the source data.

1. In Test Data Manager, click **Policies**.
2. Click **Actions** > **New** > **Masking Rule**.

   The **New Masking Rule** dialog box appears.
3. Enter a name for the masking rule.

4. Select the **Numeric** data type.
5. In the Standard masking type, select **Random**. Choose to override the masking parameters.
6. Click **Next**.
7. To configure a range of values, enter the following values:
   - Minimum. Enter 10000.
   - Maximum. Enter 80000.

   The following image shows the parameters that you can configure for a numeric random masking rule:



8. Choose to ignore the null and empty spaces. Use the default server value to mask the source data to handle errors.
9. Click **Finish**.

Step 2. Create a Shuffle Masking Rule

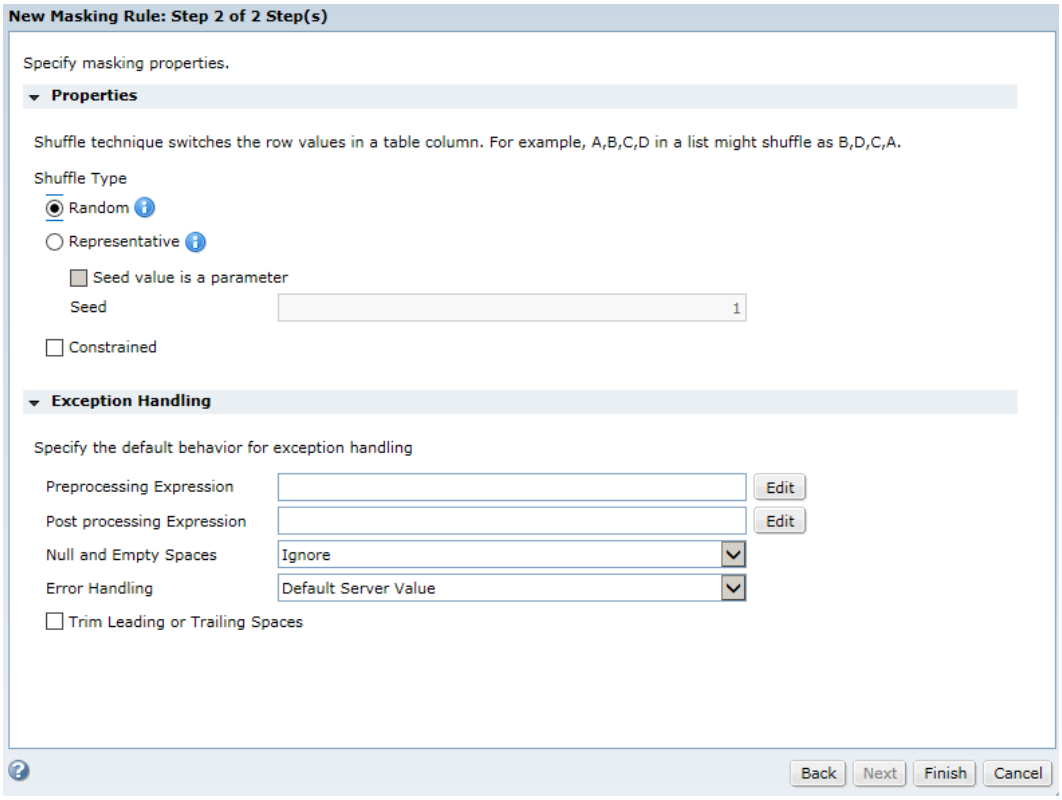
Create a shuffle masking rule to switch the row values for the column that contains the branch ID details. You can choose random shuffle type to shuffle values from one row to another without checking if the target values are unique for each source value.

1. Click **Actions** > **New** > **Masking Rule**.

   The **New Masking Rule** dialog box appears.
2. Enter a name for the masking rule.
3. Select the **String** data type.
4. In the Standard masking type, select **Shuffle**. Choose to override the masking parameters.
5. Click **Next**.

6. Choose the **Random** shuffle type.
7. Choose to ignore the null and empty spaces. Use the default server value to mask the source data to handle errors.

   The following image shows the parameters that you can configure for a shuffle masking rule:



8. Click **Finish**.

Step 3. Add the Masking Rules to a Project

After you create the numeric random and shuffle data masking rules, you must add the rules to a project. Use the random and shuffle masking rules within the project to mask target columns. In the example, the random masking rule is Balance_Mask and the shuffle masking rule is Branch_Mask.

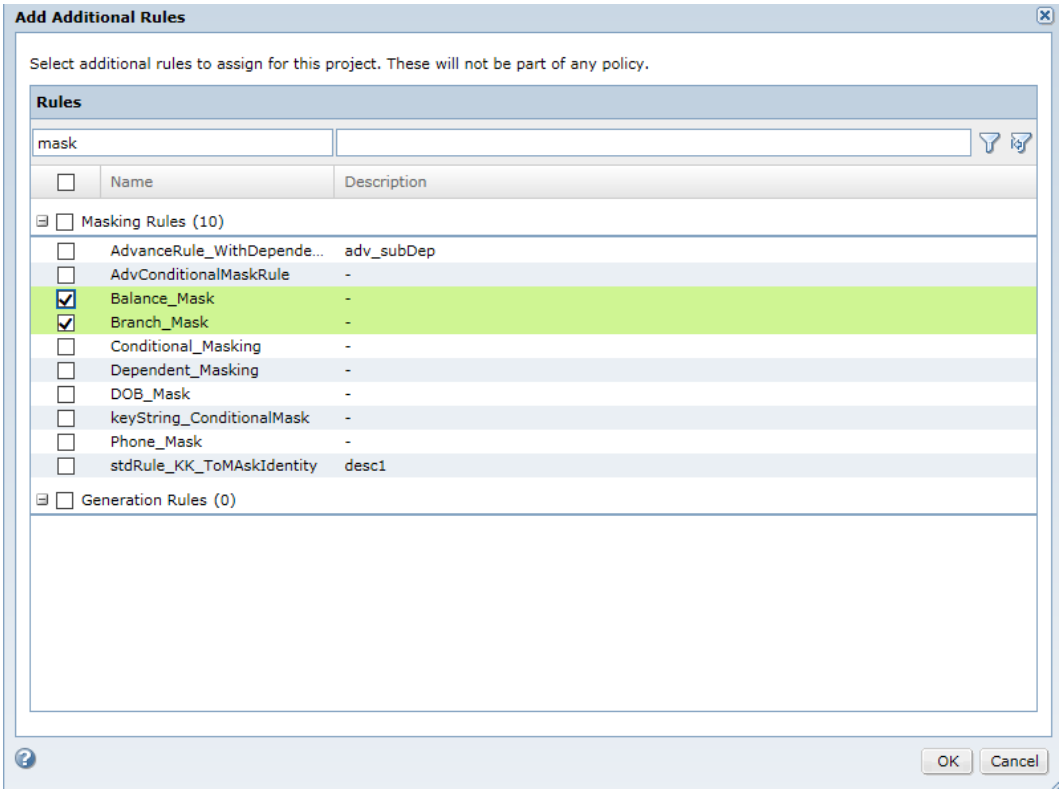
1. Click **Projects**.

   You can see a list of projects.
2. Open the project in which you want to use the rules.

   The project window opens in another tab.
3. Click **Overview** > **Policies**.
4. Click **Actions** > **Add Additional Rules**.

   The **Add Additional Rules** dialog box appears.
5. Select the numeric random and shuffle masking rules that you created.

The following image shows the masking rules that you can add to the project:



6. Click **OK**.

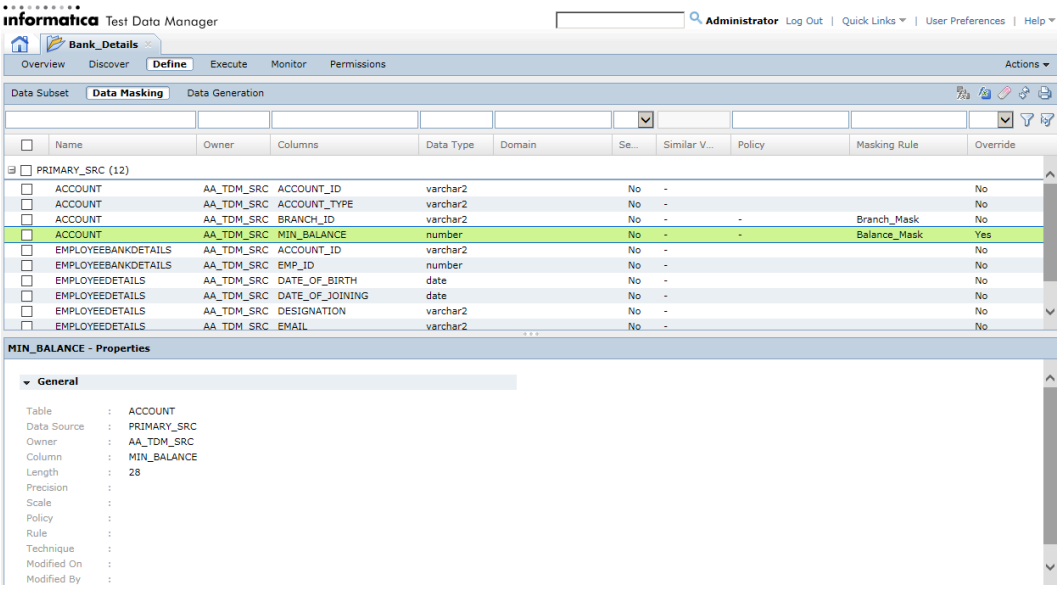
   The Balance_Mask and Branch_Mask masking rules appear under the **Additional Rules** list.

Step 4. Assign the Masking Rules to Target Columns

After you add the masking rules to the project, you can assign the data masking rules to target columns to mask sensitive data. In the ACCOUNT table, assign the numeric random masking rule to the MIN_BALANCE column to mask balance amount of the customers. Assign the shuffle masking rule to the BRANCH_ID column to mask the branch IDs the customers.

1. Click **Define** > **Data Masking**.
2. In the ACCOUNT table, select the BRANCH_ID column.
3. In the Masking Rule column, select the shuffle masking rule Branch_Mask that you created.
4. To save the rule assignment, click **Save**.
5. In the ACCOUNT table, select the MIN_BALANCE column.
6. In the Masking Rule column, select the numeric random masking rule Balance_Mask that you created.

The following image shows the masking rules assignment to the target columns:



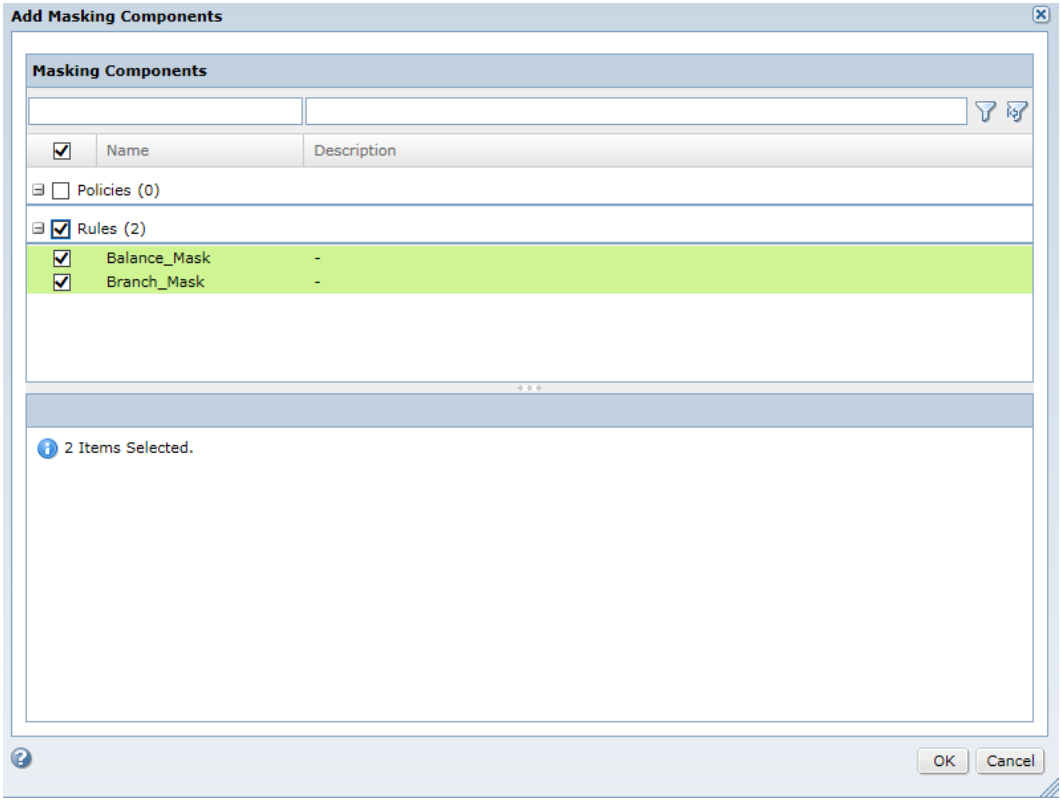
7. To save the rule assignment, click **Save**.

Step 5. Create a Data Masking Plan

Create a plan and add masking components to define the data masking operation. In the example, create a plan with the name Bank_MaskPlan. To the plan, add the Balance_Mask and Branch_Mask masking rules that you assigned to target columns. You must specify the source connection from where you can get the source data and the target connection where you want the target data with the masked columns.

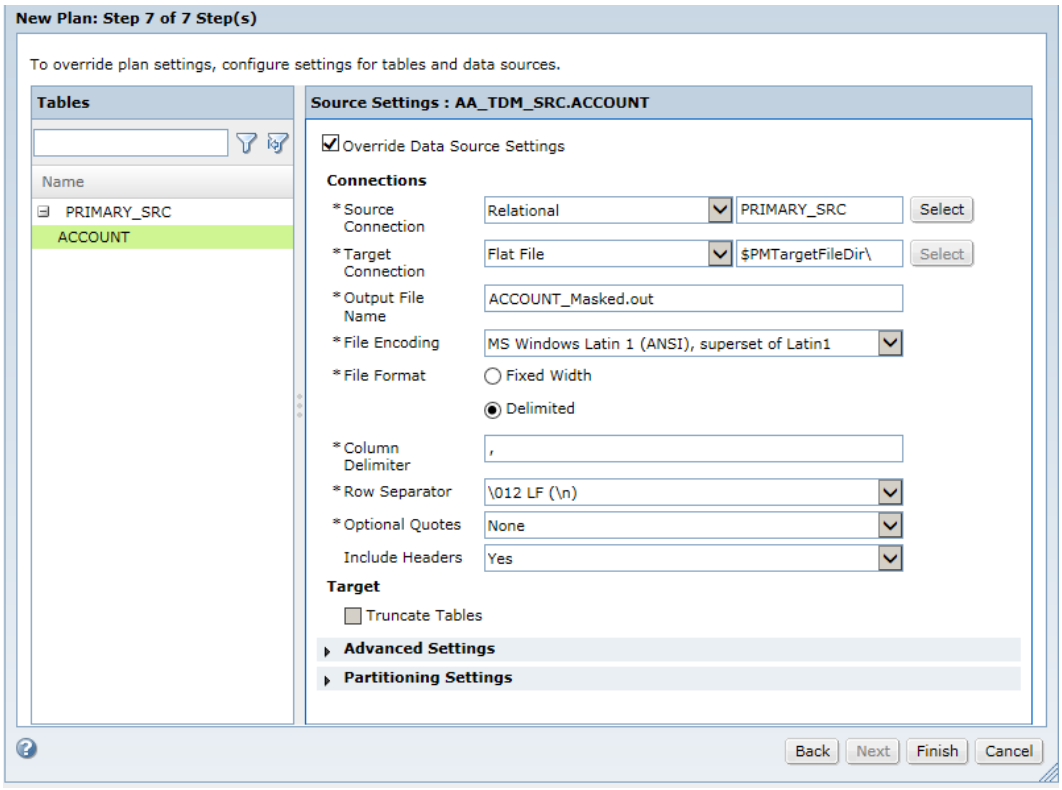
1. Click **Execute**.
2. Click **Actions** > **New**.
3. In the **New Plan** dialog box, enter a name for the plan.
4. Click **Next**.
5. To add a data masking component, click **Add Masking Components**.
6. Select both the masking rules that you created.

The following image shows the masking rules that you can add to the plan:



7. Click **OK**.
8. Click **Next**.
9. To skip adding a data subset component, click **Next**.
10. To skip adding a data generation component, click **Next**.
11. Select the relational source connection from where you imported the metadata. Select a flat file target connection where you want the target data.
12. Click **Next**.
13. Review the data source settings. To change the target flat file properties, select the **ACCOUNT** table and select **Override Data Source Settings**. Change the name of the output file.

The following image shows the source-level plan properties that you can configure for a flat file:



14. Click **Finish**.

    The plan with the name Bank_MaskPlan appears in the project.

Step 6. Generate and Run the Workflow

After you create the plan, you must run a workflow to mask data in the target. You can monitor the progress of the workflow job. When the workflow job succeeds, TDM masks data in the target columns.

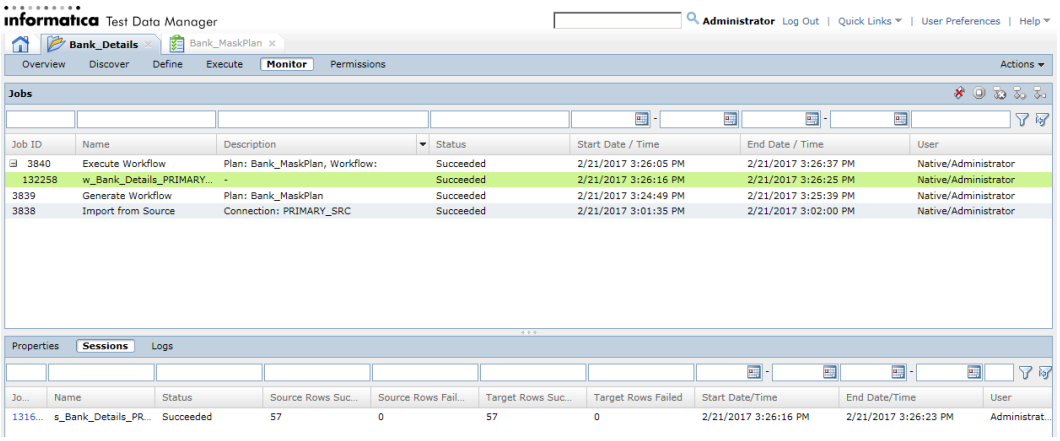
1. In the **Execute** view, click the data masking plan Bank_MaskPlan that you created.

   The plan opens in another tab.
2. Select the data masking plan that you created.
3. Click **Actions** > **Generate and Execute**.

   The **Generate and Execute Plan** dialog box appears.
4. Select the PowerCenter Integration Service.
5. Select **Schedule Now**.
6. To run the workflow, click **Execute Workflow**.
7. To view the status of the workflow job, click **Monitor**.
8. To refresh the view, click **Auto Refresh On**.

9. To view the job details, select the job ID of the plan and click the **Properties**, **Sessions**, and **Logs** tabs.

   The following image shows the status of a sample workflow job:



Sample Output

After the data masking plan succeeds, you can view that the source data is masked in the following target columns: MIN_BALANCE and BRANCH_ID. The MIN_BALANCE column contains the values between 10000 and 80000 that you defined in the numeric random masking rule. The BRANCH_ID column contains random shuffled values from another rows of the same source column.

The following image shows a sample of masked data:

```
#ACCOUNT_ID,ACCOUNT_TYPE,MIN_BALANCE,BRANCH_ID
31926819,SAVINGS,17867,BANK
7099453,CURRENT,43420,539
440116243,SALARY,22877,BANK
12300695,SAVINGS,43786,ABNA
1863000160,CURRENT,27518,762
61111101800000000,SALARY,75194,21211009
12345678,SAVINGS,33185,539
99999999,CURRENT,39867,1200
260076545510730000,SALARY,14942,200128
123456,SAVINGS,40714,10901014
0002324013AA,CURRENT,29658,BOMM0101
11101001000,SALARY,46490,200128
9400644750000,SAVINGS,41498,21211009
435195001,CURRENT,61839,AAAA
0011111000h79,SALARY,60492,500
12345678951,SAVINGS,67886,X0542811101
120000058984,CURRENT,68865,762
0012345MTLCAST001S,SALARY,48094,10800
101030300200000MUR,SAVINGS,50075,10901014
```

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